Calculating Correct and Efficient Programs

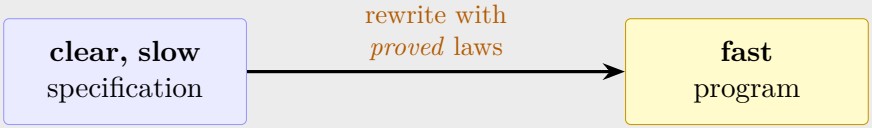
David Castro-Perez

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Based on joint work with Marco Paviotti and Michael Vollmer (University of Kent):
Program Optimisations via Hylomorphisms for Extraction of Executable Code, ITP 2025.

In One Slide



The idea (1980s):

- Programs as algebra
- Optimise by rewriting
- Correct by construction

Our work:

- Every step machine-checked
- No extra axioms
- Runnable code out

The Problem

Fast *and* Correct

A Specification and a Program



A Specification and a Program




```
let rec sort_times_two = function
  | [] -> []
  | h :: t ->
    let (l, r) = partition (fun x -> x < h) t in
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```

Are they equal? Can we *derive* the second from the first?

Three Ways to Fast and Correct Code

Approach	Good	But
Write it, then **verify** it	any program	a new proof for every program

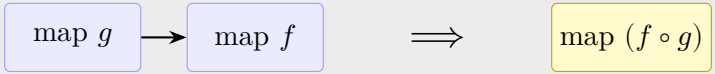
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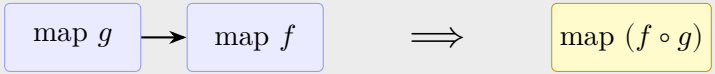
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Write it, then **verify** it	any program	a new proof for every program
Verified compiler	translation is correct	keeps your algorithm: $O(n^3)$ stays $O(n^3)$
Calculate it from a spec	proof comes with the derivation	needs laws that *apply*

You Already Trust Rewrite Rules



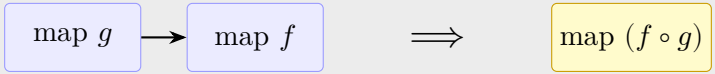
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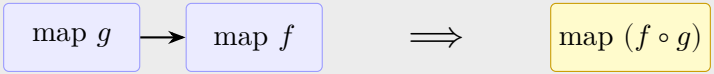
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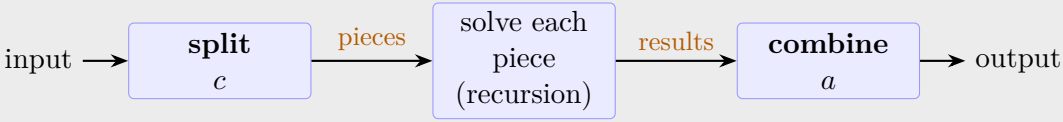


- ML compilers *fuse* tensor operations into one kernel.
- Database engines rewrite query plans.
- Compilers inline, fuse loops, remove temporary data.
- Usually: a **fixed** set of rules, **trusted**, not proved.

The Laws

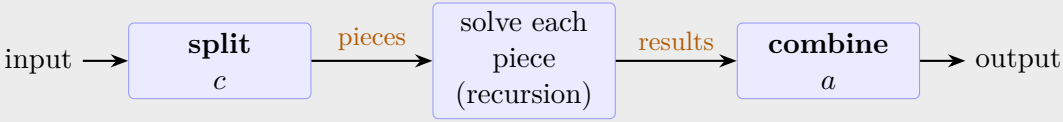
Divide and Conquer as Algebra

One Shape for Many Algorithms



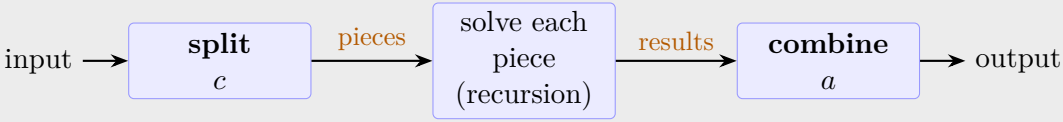
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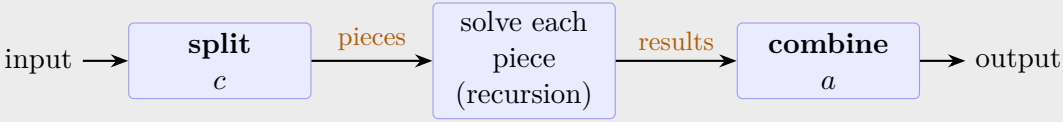
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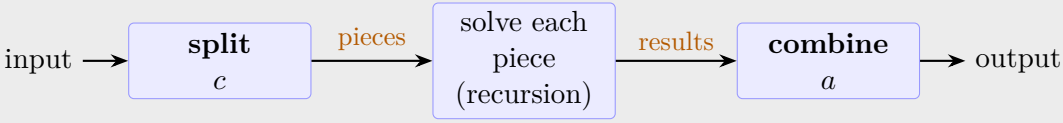
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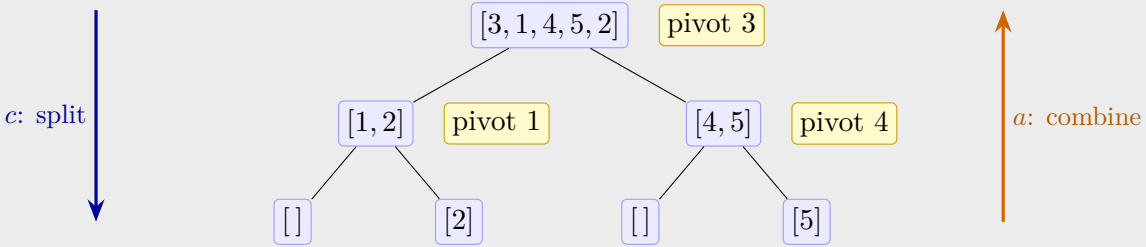
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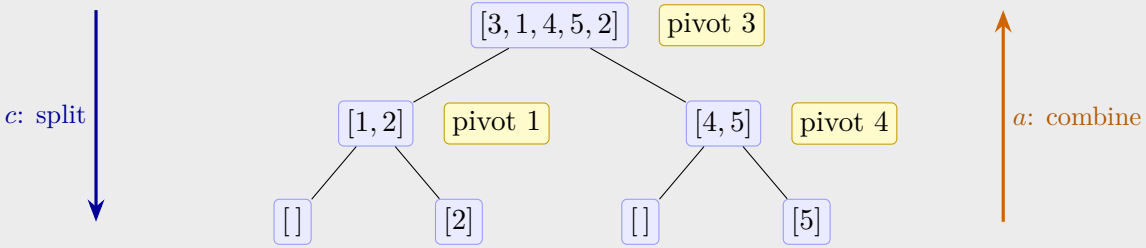
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Hylomorphism $[\![a, c]\!]$: the recursion is *generic*; only a and c change.

The Call Tree of Quicksort

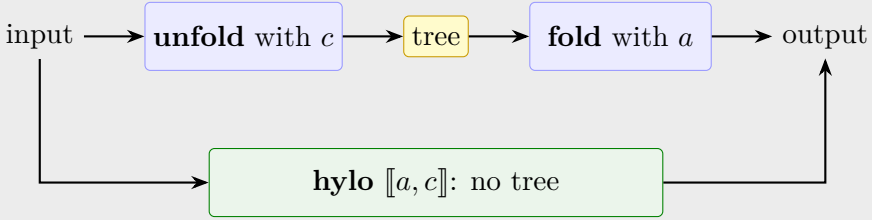


The Call Tree of Quicksort



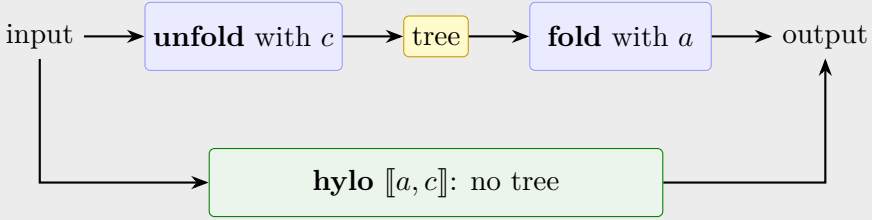
The tree is *virtual*: it is never built in memory.

Producers and Consumers



- Specification: *build*, then *consume*. Easy to read.

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- Program: do both at once. Fast.

Two Laws

Fusion $[\![a, c]\!] \rightarrow h = [\![b, c]\!]$

if h distributes over a

Deforestation $[\![\mathsf{build}, c]\!] \rightarrow [\![a, \mathsf{take_apart}]\!] = [\![a, c]\!]$

- Proved **once**, for every split, combine, data type.

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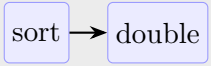
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- Proved **once**, for every split, combine, data type.
- Each use is a *rewrite step*.

Deriving the Example



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$$\mathrm{sort} \rightarrow \mathrm{double}$$

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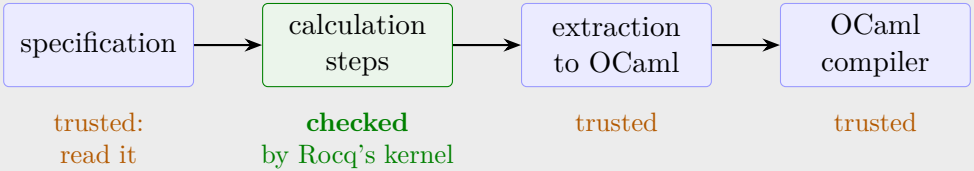
$$\begin{aligned}
&\boxed{\text{sort}} \rightarrow \boxed{\text{double}} \\
=\ &\boxed{[\![\mathsf{concat}, \mathsf{partition}]\!]} \rightarrow \boxed{\text{double}} && \text{definition of quicksort} \\
=\ &\boxed{[\![\mathsf{concat}\text{ and double pivot}, \mathsf{partition}]\!]} && \text{fusion}
\end{aligned}$$

Each $=$ is a **theorem** applied by a proof assistant.

Machine-Checked

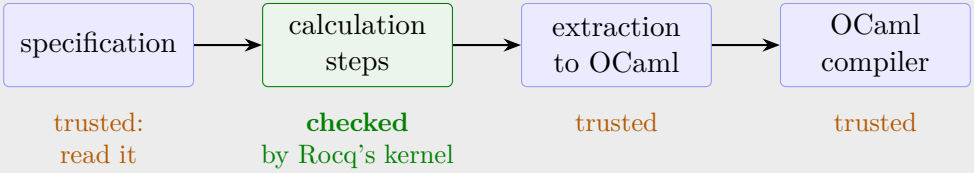
Hylomorphisms in Rocq

What You Have to Trust



- Who picked the steps does not matter: a person, a script, a model.

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- A wrong step is **rejected**, not miscompiled.

Three Obstacles

Termination

every function must provably stop

No extra axioms

comparing functions needs one

Clean code

generic proofs extract to unsafe casts

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Haskell has the laws but not termination; Rocq has termination but rejects the textbook definitions.

What the Textbook Encoding Extracts To

```
type sh = __                                    (* = Obj.t *)
type pos = __
type 'x ext = { shape : sh; cont : (pos -> 'x) }

let merge x =
  let { shape = shape0; cont = k } = x in
  (match Obj.magic shape0 with
   | Some h -> app (Obj.magic k True) (Cons (h, (Obj.magic k False)))
   | None -> Nil)
```

(simplified)

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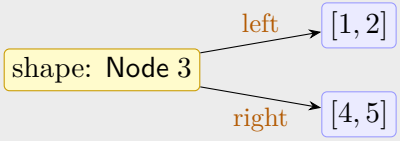
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`Obj.magic`: an **unchecked cast**. Unsafe as soon as this code meets other code.

Our Fix: Shapes and Positions

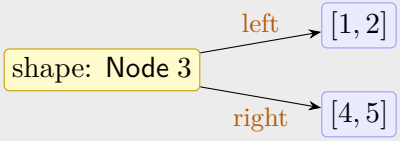


shape: Leaf no positions

valid?	left	right
`Node h`	yes	yes
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- One type of positions + a **yes/no check**: plain OCaml types.

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- One type of positions + a **yes/no check**: plain OCaml types.
- Proofs about the check are **erased** from the code.

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- The proof is scaffolding: **erased** at extraction.

No Extra Axioms

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Print Assumptions hylo_fusion_l.          Closed under the global context
Print Assumptions mss_calc.               Closed under the global context
Print Assumptions qsort_times_two_calc.   Axioms: PrimInt63.int, ltb, mul
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(simplified)

- Equality = *same results on equal inputs* (setoids).

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- Equality = *same results on equal inputs* (setoids).
- Laws become `rewrite` steps.

In Practice

From Calculation to Code

A Calculation in Rocq

```
Derive qsort_times_two
  in (Lmap times_two \o cata merge \o rana tsplit =e qsort_times_two)
  as qsort_times_two_calc.
Proof.
  rewrite map_merge_fusion, hylo_ana, deforest by apply l_out_in.
  finished.
Qed.
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specification

the laws

What Comes Out

```
let rec qsort_times_two_code = function
| [] -> []
| h :: t ->
  let (l, r) = partition (fun x0 -> ltb x0 h) t in
  let x0 = fun e ->
    qsort_times_two_code (match e with Lbranch -> l | Rbranch -> r) in
  app (x0 Lbranch) ((mul (Uint63.of_int (2)) h) :: (x0 Rbranch))
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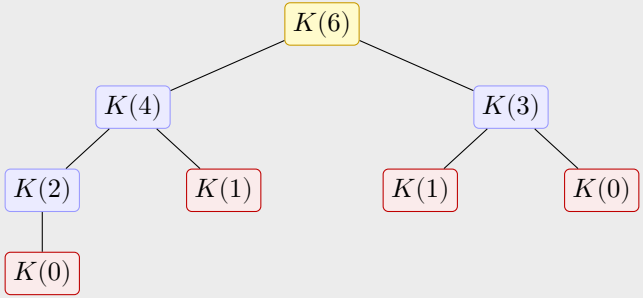
(simplified)

Hand-written (start of the talk):

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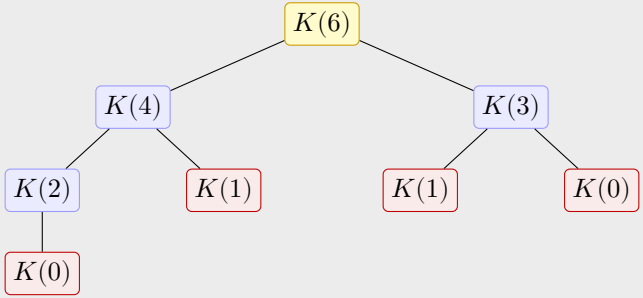
Dynamic Programming: Knapsack

$$K(0) = 0 \qquad K(c) = \max\left(\{0\} \cup \{\, K(c - w_i) + v_i \mid 0 < w_i \leq c \,\}\right)$$



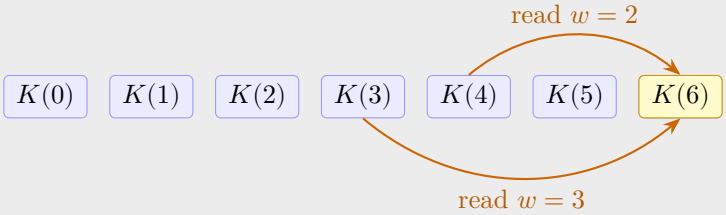
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A hylomorphism, but it **recomputes**: exponential.

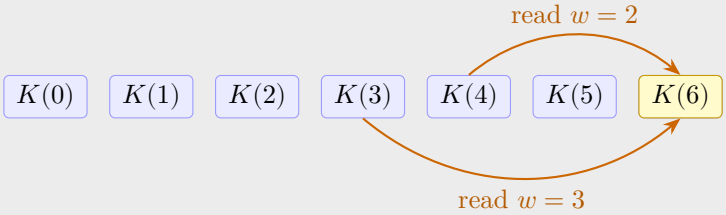
Remember, Don't Recompute



- **Dynamorphism**: a hylomorphism that keeps a *table* of earlier results.

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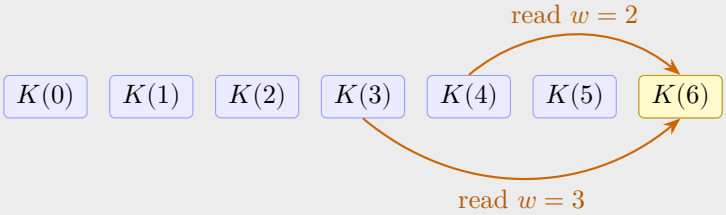
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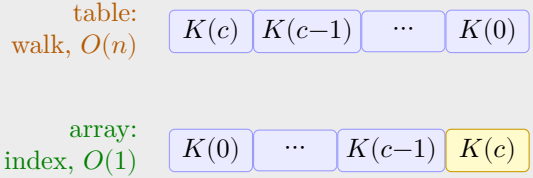
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- **Dynamorphism**: a hylomorphism that keeps a *table* of earlier results.
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- Exponential $\Longrightarrow$ one entry per capacity.

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From $O(n)$ to $O(1)$ Lookups, by Rewriting



From $O(n)$ to $O(1)$ Lookups, by Rewriting

```
(* Arrays are one more memo table: an initial algebra of Memo *)
Instance ATab_initial : Initial (Memo NatF A) (ATab A).

(* dyna does not depend on the memo table *)
Theorem dyna_iso : dyna (T := T) a c =e dyna (T := U) (a \o fmap (iso U T)) c.

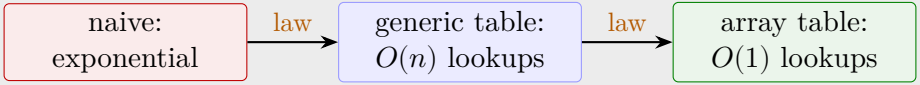
Derive knapsack_arr in (knapsack_naive wvs =e knapsack_arr) as calc.
Proof.
  rewrite <- (knapsack_correct (T := ATab int)).  (* naive hylo = dyna *)
  unfold knapsack; rewrite dyna_memo. finished.   (* share the table *)
Qed.
```

(simplified)

Knapsack: the Extracted Program

```
let rec memo t c = function
  | [] -> []
  | (w, v) :: wvs ->
    if 0 < w && w <= c then (get t (c - w) + v) :: memo t c wvs
    else memo t c wvs
let rec table wvs c =
  if c = 0 then make 1 0
  else let t = table wvs (c - 1) in
       push t (max_int 0 (memo t c wvs))
let knapsack wvs c = newest (table wvs c)
```

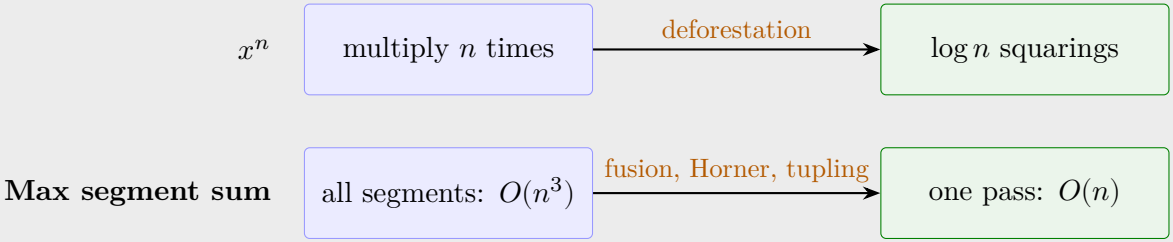
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More Calculations

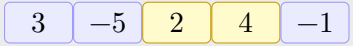


More Calculations



Maximum Segment Sum: Spec and Result

$$\mathsf{mss} = \mathsf{maximum} \circ \mathsf{map}\,\mathsf{sum} \circ \mathsf{segments}$$



best segment: $2 + 4 = 6$

Maximum Segment Sum: Spec and Result

$$\text{mss} = \text{maximum} \circ \text{map}\,\text{sum} \circ \text{segments}$$



best segment: $2 + 4 = 6$

```
let mss_code xs =
  let rec f = function
  | [] -> (Some 0, Some 0)
  | h :: t ->
    let (m, s) = f t in
    let s' = match s with
      | Some b -> Some (max 0 (h + b))
      | None -> Some 0 in
    (omax s' m, s')
  in fst (f xs)
```

(simplified)

Summary So Far

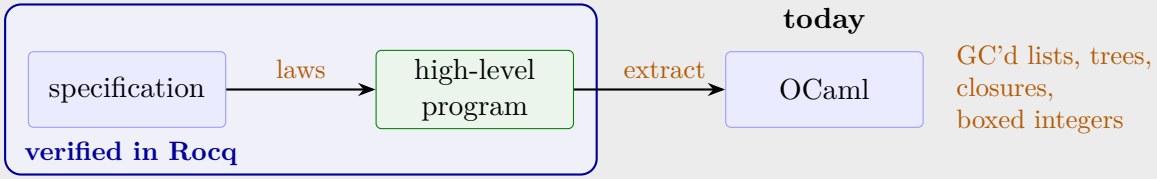
- A **library**: hylomorphisms and their laws in Rocq.
- Optimisations are **proofs by rewriting**.
- Termination proved **once per split**.
- **No extra axioms**; OCaml **without unsafe casts**.

D. Castro-Perez, M. Paviotti, M. Vollmer. **Program Optimisations via Hylomorphisms for Extraction of Executable Code**. ITP 2025. Library: `https://github.com/dcastrop/rocq-hylomorphisms`

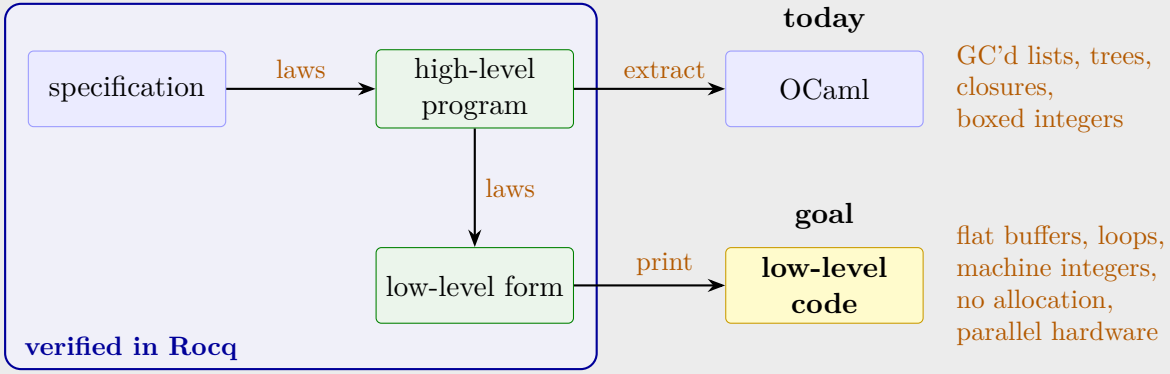
Work in Progress

Down to Low-Level Code

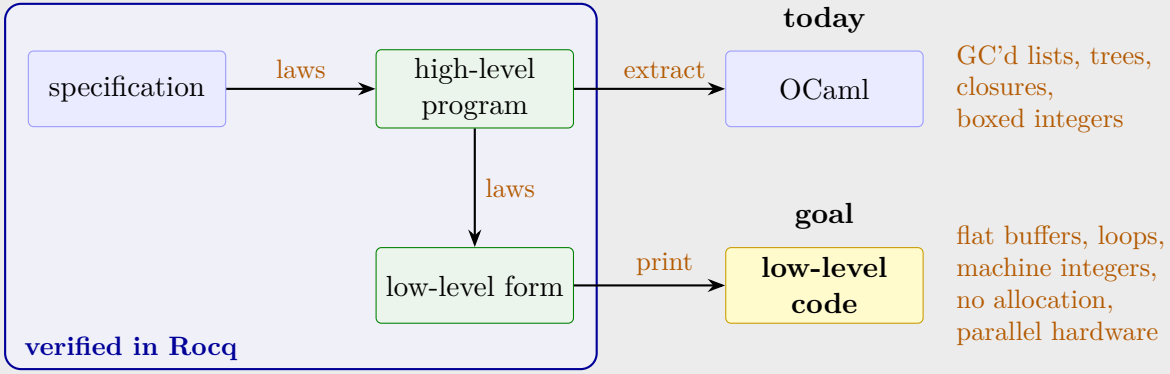
Two Ways Out of Rocq



Two Ways Out of Rocq



Two Ways Out of Rocq



Not "OCaml, then lower": **more calculation** reaches the low-level form; extraction is a direct print.

What We Are Working On

Low-level, **correct-by-construction** code,
extracted from **high-level calculation steps**,
possibly guided by a **cost model**.

- You choose *laws*, not pointers and indices.

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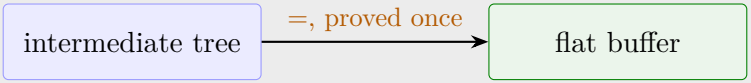
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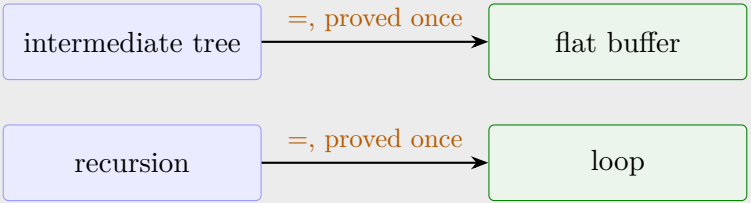
Low-level, **correct-by-construction** code,
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possibly guided by a **cost model**.

- You choose *laws*, not pointers and indices.
- Every step proved; the output is machine-level code.
- A cost model says *which* equal program to pick.

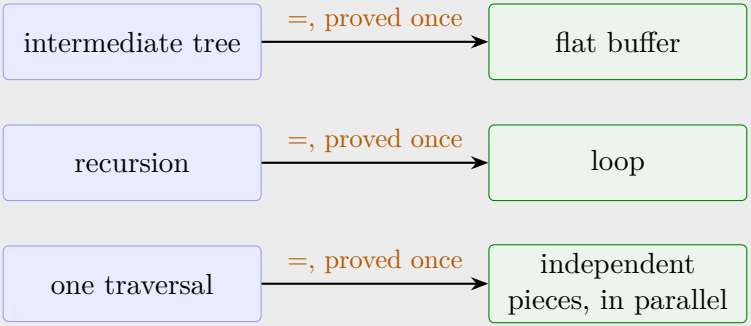
Ongoing: Laws That Change Representation



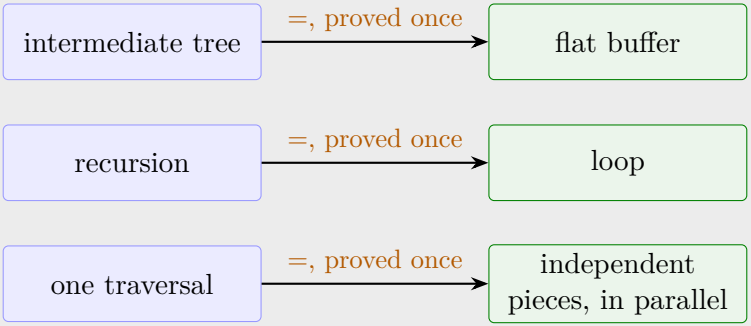
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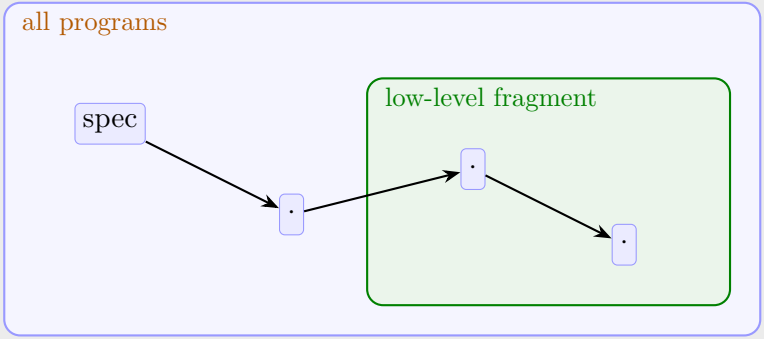


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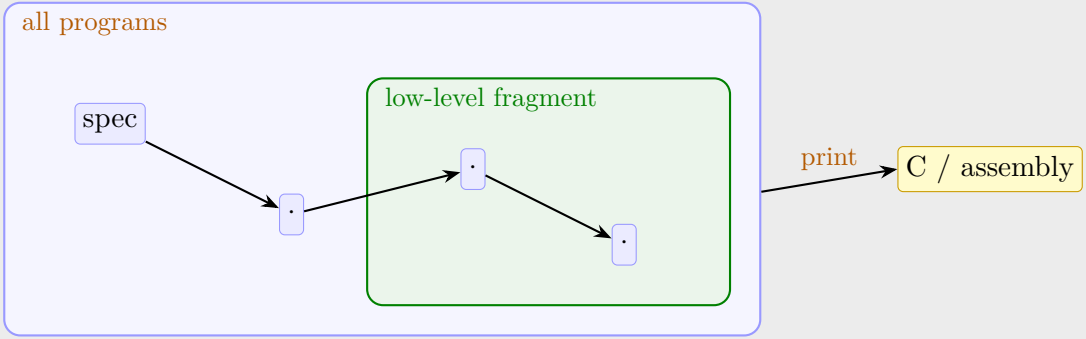
Today's laws change the *algorithm*; these change *where data lives* and *how work is scheduled*.

Ongoing: A Target the Laws Never Leave



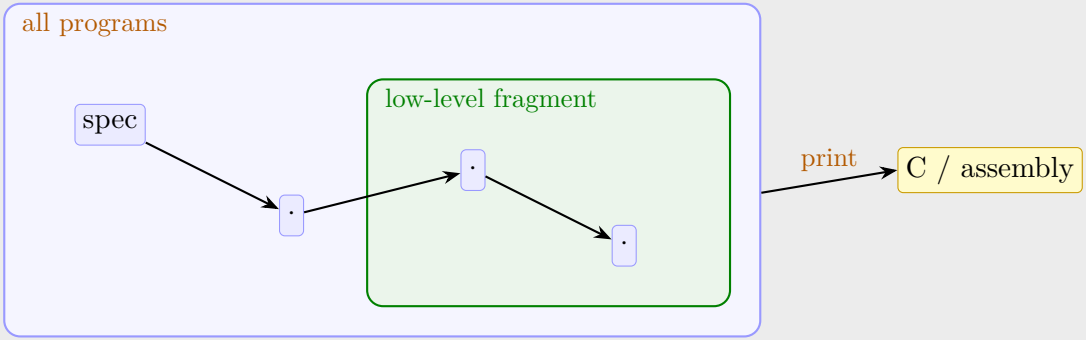
- Every law keeps programs **inside** the fragment.

Ongoing: A Target the Laws Never Leave



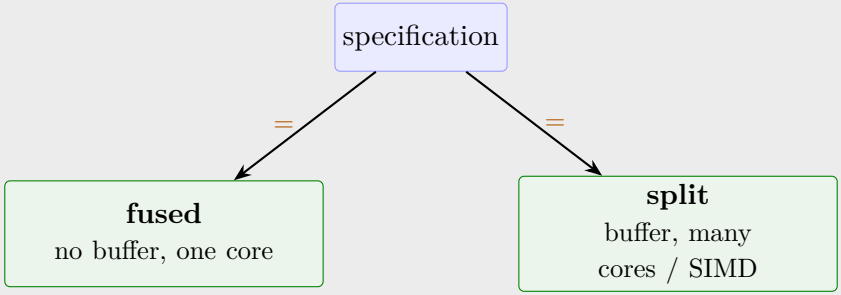
- Every law keeps programs **inside** the fragment.
- Only a **small printer** is trusted.

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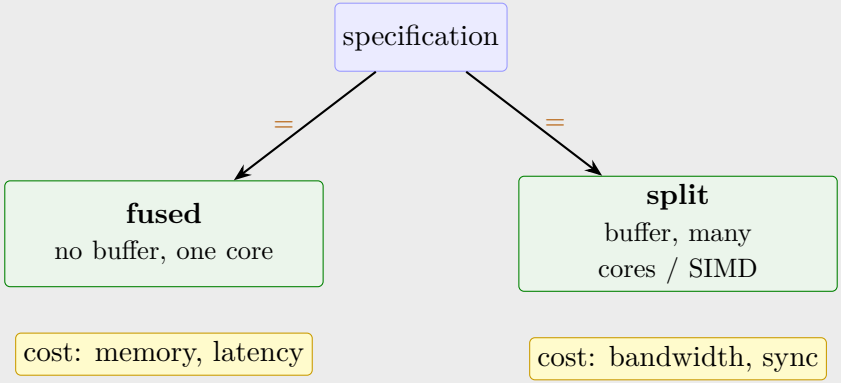


- Every law keeps programs **inside** the fragment.
- Only a **small printer** is trusted.
- Overflow and bounds: **side conditions** of the laws.

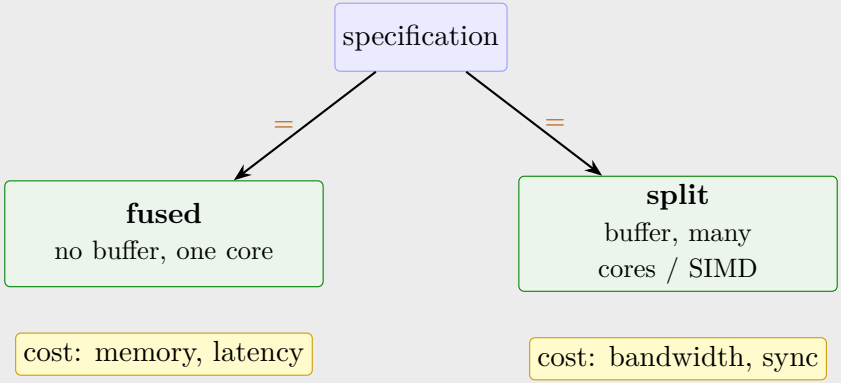
Ongoing: A Cost Model



Ongoing: A Cost Model

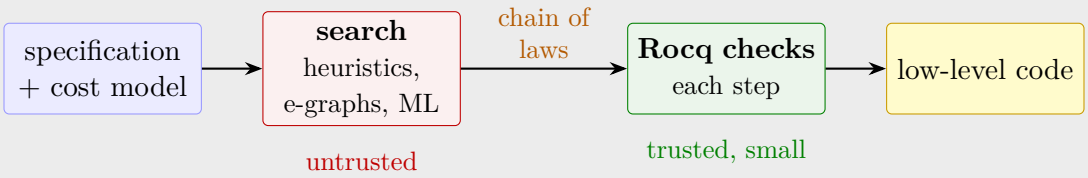


Ongoing: A Cost Model



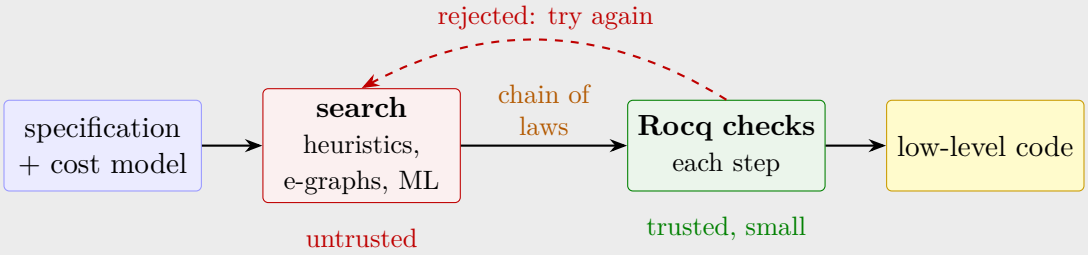
The laws say what is *equal*; a cost model says what is *better*.

Untrusted Search, Trusted Checking



- The derivation is a **certificate**.

Untrusted Search, Trusted Checking



- The derivation is a **certificate**.
- Search can be as clever, or as wrong, as it likes.

Where We Are



In Rocq, axiom-free:

- Splitting index ranges k ways
- Machine integers, no overflow (proved)
- Buffers, compared by their reads
- Tree reductions over ranges
- Writes to a buffer, driven by any split

Not yet:

- Laws that add or remove a buffer
- Each index written exactly once
- The low-level target and printer
- The cost model and search

Summary

- Program optimisation as **algebra**: rewrite with laws.
- Divide and conquer as **hylomorphisms**: laws proved once.
- In Rocq: **machine-checked**, no extra axioms, clean OCaml.
- Ongoing: **low-level, correct-by-construction code** from high-level steps, guided by a **cost model**.

Thank you!

Questions?

`https://github.com/dcastrop/rocq-hylomorphisms`

Based on joint work with Marco Paviotti and Michael Vollmer.